ELSEWEDY UNIVERSITY OF TECHNOLOGY
- POLYTECHNIC OF EGYPT -

smart industrial Destribution board

· Supervised by: Dr. mohamed mandour

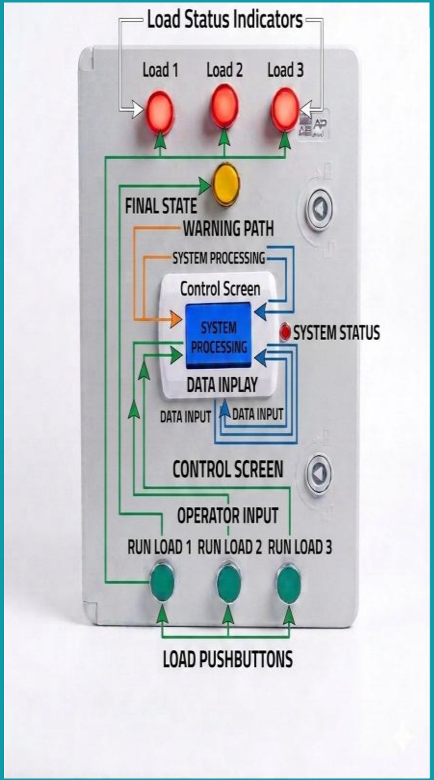
[Department] · Faculty of electrical Engineering technology

Semester: Spring 2025 — 2nd Year

PROJECT OVERVIEW

We built an Arduino-based Smart Distribution Board providing 3-in-1 power protection with a black-box failure logger. It eliminates sensor noise and display flickering, offering a cost-effective alternative to expensive commercial equipment. This directly benefits maintenance technicians by simplifying troubleshooting and minimizing factory downtime.

… 8 member	… 14Weeks	embedded system
Team Size	Duration	Tech Stack



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01 # TEAM MEMBERS

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proplem solving

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Supervised by: Dr. Supervisor Name | Co-Supervisor / TA: Eng. Name

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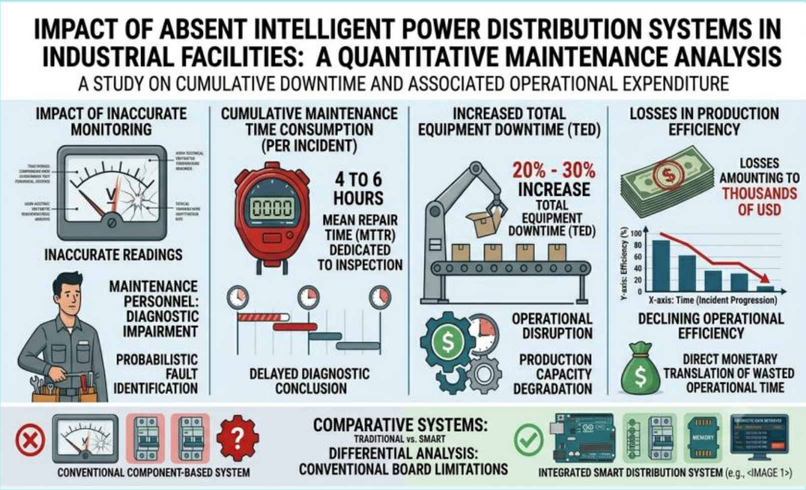
PROBLEM STATEMENT

THE PROBLEM

Industrial distribution boards frequently suffer from high sensor noise and rapid screen flickering, making real-time electrical parameters difficult to read accurately. Additionally, when critical power spikes or voltage drops cause sudden system failures, vital diagnostic data is instantly lost, leaving technicians with no way to trace the root cause. This project addresses these stability and data-loss challenges by introducing hardware-software filtering and an EEPROM-based "black box" logger to ensure reliable monitoring and instant troubleshooting for maintenance personnel.



WHY IT MATTERS

Impact 1: Unstable sensor readings and the instant loss of diagnostic data during power failures blind maintenance teams, resulting in prolonged equipment downtime and expensive, trial-and-error repairs.

Impact 2: **Maintenance technicians and plant operators are directly affected. Unstable displays and missing failure data leave them blind to the root causes of trips, forcing them into slow, expensive trial-and-error repairs that prolong costly factory downtime.**

Impact 3: In industrial plants, unstable monitoring and unlogged power faults force maintenance teams to waste up to 4 to 6 hours per breakdown on blind troubleshooting. For a typical manufacturing line, this diagnostic delay contributes to a 20% to 30% increase in total equipment downtime, translating directly into thousands of dollars in lost production efficiency.

75%

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PROPOSED SOLUTION

Summarize your solution in one sentence — what it does, for whom, and why it works.

microcontroller

Processes software filtering to eliminate display flickering and commands rapid fault tripping to protect loads.

EEPROM

Saves critical diagnostic data at the exact moment of power failure to eliminate troubleshooting blindness.

smart sensors & filtring stage

aptures electrical data while actively isolating sensor noise to deliver stable, real-time metrics.

SYSTEM ARCHITECTURE & METHODOLOGY

1

Data Collection

Tools, datasets, or sources used to gather input.

2

Preprocessing

Cleaning, formatting, and transforming raw data.

3

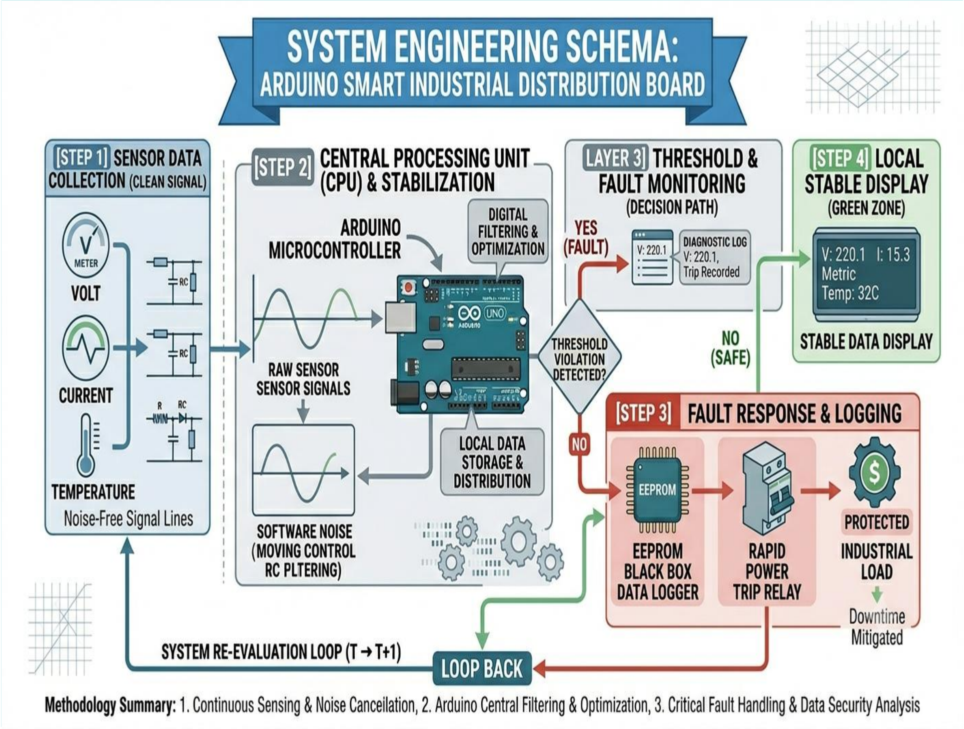
Core Logic / Model

Algorithm, model, or main processing component.

4

Output & Interface

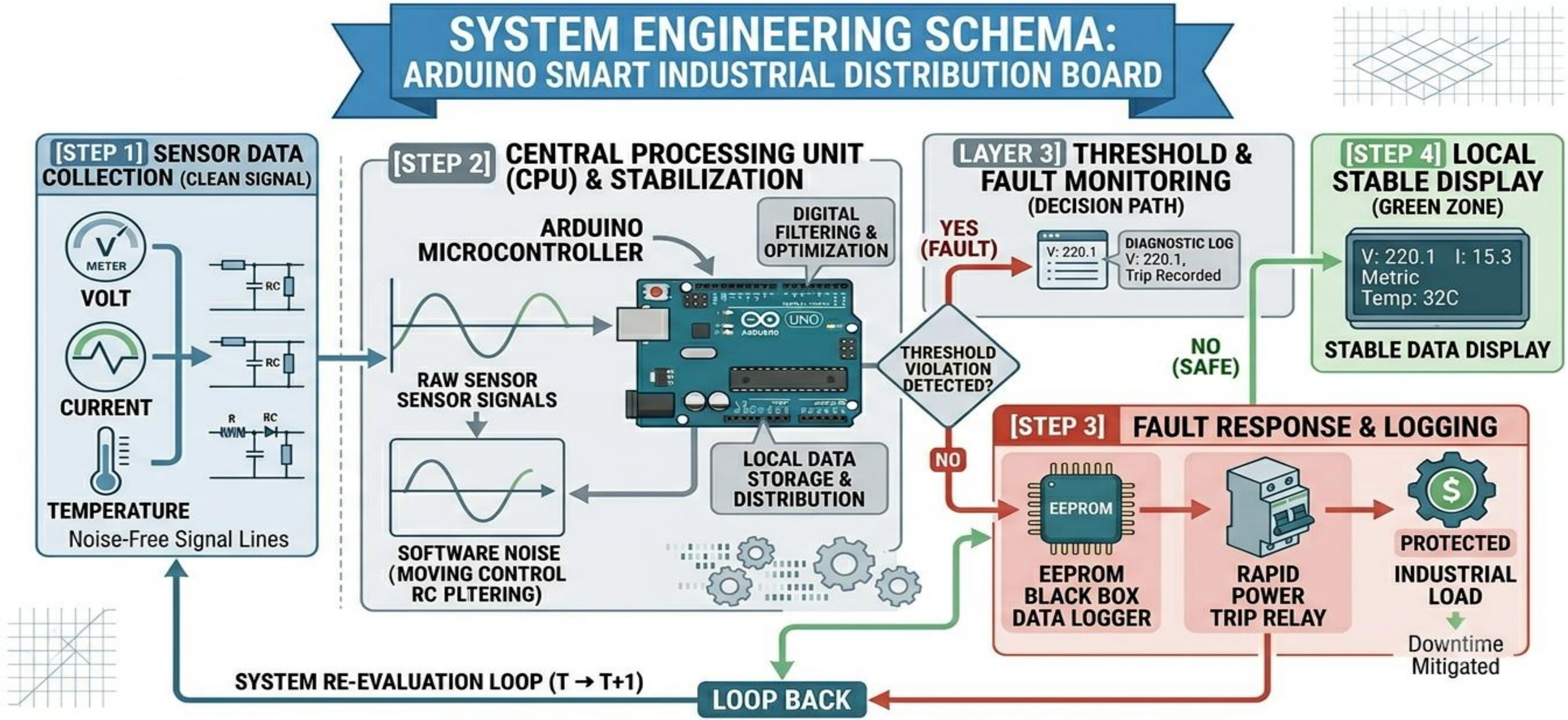
How results are presented to users.



TOOLS & TECHNOLOGIES

Arduino IDE (C/C++)

SYSTEM ARCHITECTURE & METHODOLOGY



Methodology Summary: 1. Continuous Sensing & Noise Canceilation, 2. Arduino Central Filtering & Optimization, 3. Critical Fault Handling & Data Security Analysis

DEMO & SCREENSHOTS

Fig. 1 — Main interface / key feature demonstration

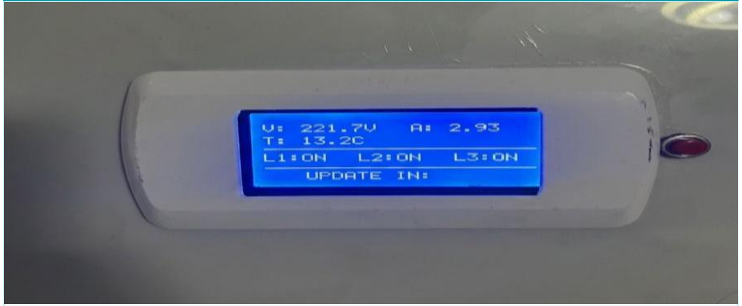
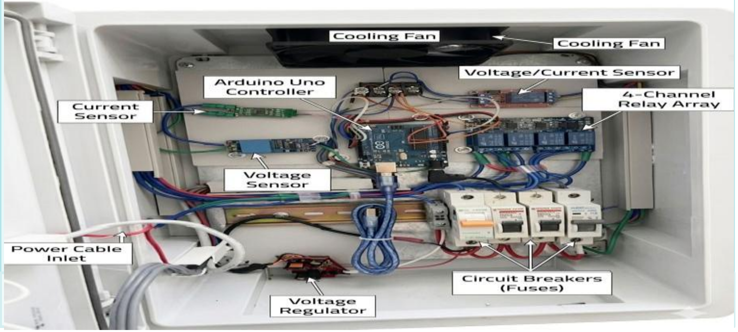


Fig. 2 — Feature / screen description

Figure 4-1: Internal Wiring Diagram of the Control System



⬡ Live demo available during presentation

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RESULTS & EVALUATION

91.3%

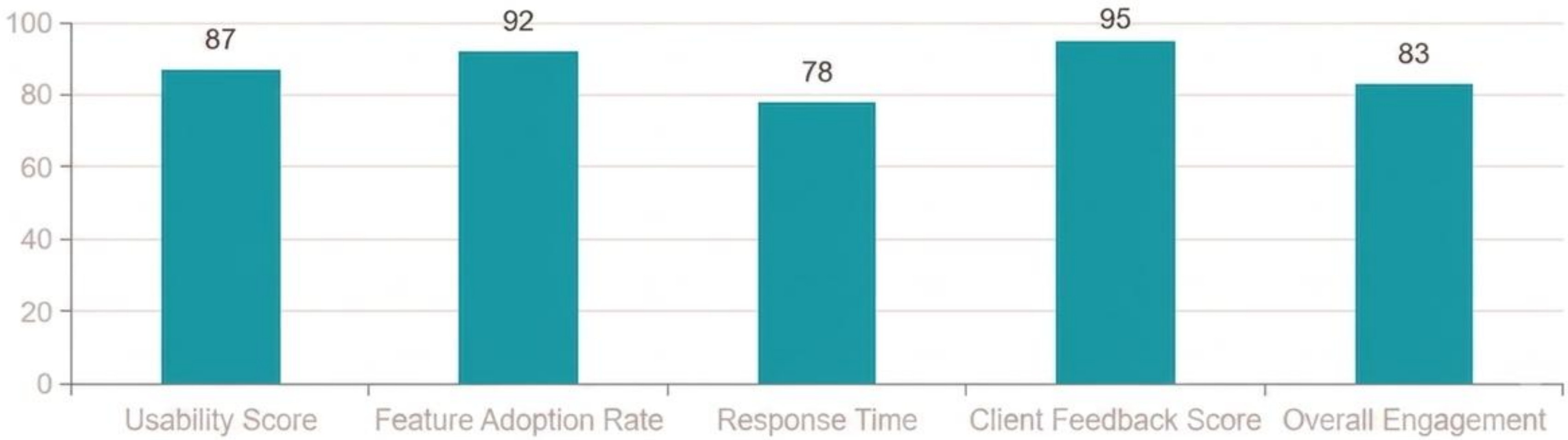
Accuracy / Score

0.94

Performance Metric

4.7/5

User Satisfaction



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PROJECT TIMELINE

Phase 1	Phase 2	Phase 3	Phase 4	Phase 5
Research & Requirements	**Design & Prototyping**	**Development Sprint**	**Testing & Evaluation**	**Final Report & Demo**
Wk 1–2	Wk 3–4	Wk 5–8	Wk 9–10	Wk 11–12
Requirements Document	Design Prototype	Working System	Test Report	Final Presentation

Thank You

We welcome your questions

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